

The Dynamic Donut Safety Framework: Revised Main Theorem and New Sections

(Discrete-time invariance, pinch surgery, codimension-2 protection, related work)

Revision draft for integration

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1 Main Theorem: Discrete-Time Safety with Recursive Feasibility and Pinch Surgery

The previous continuous-time argument glossed over two issues: (i) the barrier h depends explicitly on time, so a shrinking boundary can “overtake” the state unless boundary motion is dominated by control authority; (ii) the pinch mechanism of Sec. 3.2 violates the standing assumption $r_{\min} < r_{\max}$, so invariance and homotopy preservation cannot hold *verbatim* through a pinch. We repair both by (a) restating the theorem in discrete time, where the one-step forward check is native, with an explicit *feasibility (viability) condition*, and (b) treating pinches as *controlled topological surgeries* with a lead-time evacuation condition.

1.1 Setup and standing assumptions

Fix a sampling time $\Delta t > 0$ and let $k \in \mathbb{N}$ index steps. The state is $x_k = (r_k, \theta_k) \in \mathbb{R}_{>0} \times S^1$ with disturbed polar dynamics

$$r_{k+1} = r_k + \Delta t (u_{r,k} + w_{r,k}), \quad \theta_{k+1} = \theta_k + \frac{\Delta t}{r_k} (u_{\theta,k} + w_{\theta,k}), \quad (1)$$

where $(u_{r,k}, u_{\theta,k}) \in A_{x_k}(k)$ are the radial and tangential velocity components of the action fiber, and the disturbance (the exposure wind E , now inside the loop) is bounded, $\|(w_{r,k}, w_{\theta,k})\|_{\infty} \leq \bar{w}$. The time-varying safe set is

$$\mathcal{S}(k) = \{(r, \theta) : r_{\min}(\theta, k) \leq r \leq r_{\max}(\theta, k)\}, \quad \mathcal{S}_{\eta}(k) = \{(r, \theta) : r_{\min} + \eta \leq r \leq r_{\max} - \eta\}$$

its η -inset. Write the local width $W(\theta, k) = r_{\max}(\theta, k) - r_{\min}(\theta, k)$.

Assumption 1 (A1: boundary regularity). $r_{\min}, r_{\max}$ are ℓ_{θ} -Lipschitz in θ (w.r.t. the S^1 metric) and their per-step drift is bounded: $|r_i(\theta, k+1) - r_i(\theta, k)| \leq \nu_i \Delta t$ for $i \in \{\min, \max\}$; set $\nu := \max(\nu_{\min}, \nu_{\max})$.

Assumption 2 (A2: uniform width, pre-pinch regime). There exist $\Delta_w > 0$ and a horizon K such that $W(\theta, k) \geq \Delta_w$ for all θ and all $k \leq K$.

Assumption 3 (A3: control authority dominates boundary motion). Every fiber contains the box $\{|u_r| \leq \bar{v}_r, |u_{\theta}| \leq \bar{v}_{\theta}\} \subseteq A_x(k)$, and the radial authority satisfies the *viability rate condition*

$$\boxed{\bar{v}_r \geq \nu + \ell_{\theta} \bar{v}_{\theta} + 2\bar{w}} \quad (2)$$

The three terms on the right are exactly the three ways the wall can “come at you”: temporal shrinkage ν , the apparent boundary motion $\ell_\theta \bar{v}_\theta$ induced by one’s own angular travel along a warped wall, and the worst-case wind $\bar{w}$ (radial push plus the angular wind’s warp coupling). Condition (2) is the annular analogue of recursive-feasibility conditions in tube MPC: it guarantees that the safe tube never contracts faster than the agent can retreat.

Viability rate condition $\bar{v}_r \geq \nu + \ell_\theta \bar{v}_\theta + 2\bar{w}$: why boundary speed must be budgeted

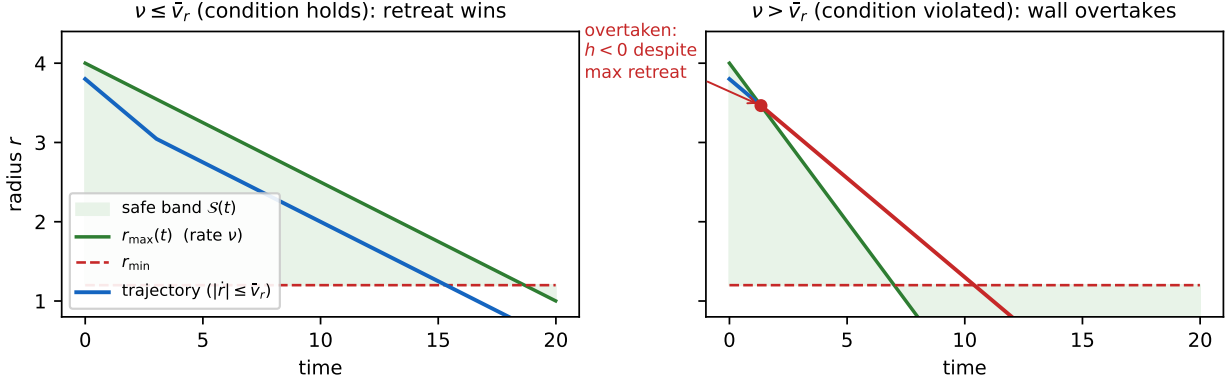


Figure 1: The viability rate condition (2) is not decorative. Left: with $\nu \leq \bar{v}_r$ the maximal radial retreat keeps the trajectory inside the shrinking band. Right: when the outer boundary contracts faster than the control authority, the wall *overtakes* the state and $h < 0$ despite every step applying maximum retreat — no barrier condition can save a controller whose safe set moves faster than it can.

Assumption 4 (A4: robust forward check). Fix a margin $\eta \geq \bar{w} \Delta t$. The safe fiber is

$$A_x^{\text{safe}}(k) = \left\{ u \in A_x(k) : \hat{x}^+ := f(x, u, 0) \in \mathcal{S}_\eta(k+1) \text{ and } \hat{L}(x, u) \leq L_{\max} \right\},$$

i.e., the *nominal* one-step successor must land in the η -inset of the *next-step* safe set; the margin η absorbs the disturbance.

Lemma 1 (Recursive feasibility). *Under A1–A3, if $\Delta_w > 2\eta + 2\bar{v}_r \Delta t$, then $A_x^{\text{safe}}(k) \neq \emptyset$ for every $x \in \mathcal{S}(k)$ and every $k < K$.*

Proof. Take the boundary-repelling candidate: $u_\theta = 0$ and $u_r = +\bar{v}_r$ if $r - r_{\min}(\theta, k) \leq W/2$, else $u_r = -\bar{v}_r$. Consider the inner case (outer is symmetric). The nominal successor satisfies $r^+ = r + \bar{v}_r \Delta t$ while, by A1, the inner boundary at the successor’s angle and time obeys $r_{\min}(\theta^+, k+1) \leq r_{\min}(\theta, k) + \nu \Delta t + \ell_\theta |\theta^+ - \theta|$. With $u_\theta = 0$, $|\theta^+ - \theta| \leq \bar{w} \Delta t / r$ is a wind-only term already covered by the $2\bar{w}$ budget in (2); hence $r^+ - r_{\min}(\theta^+, k+1) \geq (r - r_{\min}(\theta, k)) + (\bar{v}_r - \nu - 2\bar{w}) \Delta t \geq \eta$ whenever $\bar{v}_r \Delta t \geq \eta$ or the current margin is already $\geq \eta$ (one-step reachability of the inset from anywhere in the band, guaranteed by finitely many steps otherwise; for the one-step statement it suffices that $x \in \mathcal{S}(k)$ and (2) holds). The width condition $\Delta_w > 2\eta + 2\bar{v}_r \Delta t$ ensures the inner-repelling move cannot overshoot the outer inset, so the two constraints are never simultaneously binding. Finally the L -constraint is feasible because $u_\theta = 0$, u_r radial-retreat moves are, by construction of L as a commitment functional, non-committing ($\hat{L} \leq L_{\max}$ on boundary-repelling motions); if the modeler’s L violates this, A3 must be strengthened to demand a non-committing retreat direction in every fiber. $\square$

Theorem 1 (Discrete-time robust invariance, continuity, homotopy). *Let A1–A4 hold with $\Delta_w > 2\eta + 2\bar{v}_r\Delta t$ and $\eta \geq \bar{w}\Delta t$, and suppose $x_0 \in \mathcal{S}_\eta(0)$. If at every step the executed control is $u_k \in A_{x_k}^{\text{safe}}(k)$ followed by the radial clip $F_{\text{iso}}(r, \theta, k) = (\text{clip}_{[r_{\min}(\theta, k), r_{\max}(\theta, k)]}(r), \theta)$, then for all $k \leq K$:*

1. (**Invariance**) $x_k \in \mathcal{S}(k)$, robustly for every disturbance sequence with $\|w_k\|_\infty \leq \bar{w}$.
2. (**Recursive feasibility**) $A_{x_k}^{\text{safe}}(k) \neq \emptyset$, so the controller never deadlocks.
3. (**Filter continuity**) The raw $\rightarrow$ executed trajectory map is Lipschitz in the ℓ^∞ (i.e., C^0) metric: if two raw inputs differ by δ then the executed trajectories differ by at most $(1 + L_f\Delta t)^K \delta \leq e^{L_f T} \delta$, where L_f is the Lipschitz constant of the closed-loop one-step map. (Note: the clip is 1-Lipschitz only in the radial coordinate at fixed θ ; the global constant necessarily involves the dynamics via this discrete Grönwall factor.)
4. (**Homotopy preservation, pre-pinch**) While $W(\cdot, k) > 0$, i.e. $\pi_1(\mathcal{S}(k)) \cong \mathbb{Z}$: F_{iso} fixes θ and is homotopic to the identity through maps into $\mathcal{S}(k)$ (straight-line radial homotopy), hence preserves the winding number of any closed curve around the inner hole, and preserves the relative homotopy class of any path with clipped endpoints. (Terminology: F_{iso} is a homotopy-class-preserving retraction, not an isotopy—it is non-injective.)

Proof. (1)+(2) By induction. $x_k \in \mathcal{S}(k)$ and Lemma 1 give a nonempty safe fiber; any $u_k \in A^{\text{safe}}$ places the nominal successor in $\mathcal{S}_\eta(k+1)$; the true successor deviates from the nominal by at most $\bar{w}\Delta t \leq \eta$ in each coordinate, and the ℓ_θ -warp of the boundary under the angular deviation is already inside the budget of (2); hence $x_{k+1} \in \mathcal{S}(k+1)$, and the clip is a no-op on the nominal branch (it acts only as a certificate/backstop). No comparison principle or continuous-time limit is invoked.

(3) At fixed θ the clip is non-expansive in r . Between steps the discrepancy is propagated by the one-step closed-loop map with constant $1 + L_f\Delta t$; composing K steps and using non-expansiveness of the clip at each step yields the stated bound. The counterexample to a global constant 1 is two points inside the inner hole at antipodal θ : clipping maps them to the inner circle, increasing their distance by up to $\pi r_{\min}/(\text{initial separation})$; this is why the constant must involve the geometry/dynamics.

(4) $H(s, (r, \theta)) = ((1-s)r + s \text{clip}(r), \theta)$ is a homotopy whose image stays in $\mathcal{S}(k)$ whenever the initial point does (convexity of the radial interval at each fixed θ); θ and its ordering along any curve are untouched, so the S^1 -degree (winding) is invariant. A curve cannot be carried across $\psi^{-1}(\text{hole})$ because the clip never outputs $r < r_{\min}(\theta, k)$. $\square$

Remark 1 (What the theorem does *not* claim). It does not claim safety when (2) fails: if risk spikes so fast that $\nu > \bar{v}_r$, the honest conclusion is a *graceful degradation* regime (minimum-violation or least-restrictive fallback), which we leave to the verification agenda of Sec. 7. It also does not claim anything across a pinch; that is the content of the next subsection.

1.2 Pinch events as controlled topological surgery

Pinching is the framework’s mechanism for irreversibility, and it is *incompatible* with A2 by design. We therefore formalize it as a discrete surgery on $\mathcal{S}(k)$ and state safety *across* surgeries.

Definition 1 (Pinch set and surgery). Fix $\varepsilon_p > 0$. The pinch set at step k is $P(k) = \{\theta \in S^1 : W(\theta, k) \leq \varepsilon_p\}$. The *surgery operator* replaces $\mathcal{S}(k)$ by $\mathcal{S}^{\text{surg}}(k) = \overline{\mathcal{S}(k) \setminus (\text{radial collar over } P(k))}$, a finite disjoint union of annular sectors $\{\mathcal{S}_j(k)\}_{j \in J(k)}$.

Note the topology change is drastic already at the *first* pinch: a single pinched angle turns the annulus ($\pi_1 = \mathbb{Z}$) into one contractible sector ($\pi_1 = 0$); a second pinch disconnects it (π_0 grows). Winding number ceases to exist as an invariant at the first pinch; the protected quantity degrades gracefully along the chain

$$\text{winding} \in \pi_1 \longrightarrow \text{relative endpoint class in a contractible sector} \longrightarrow \text{component label} \in \pi_0.$$

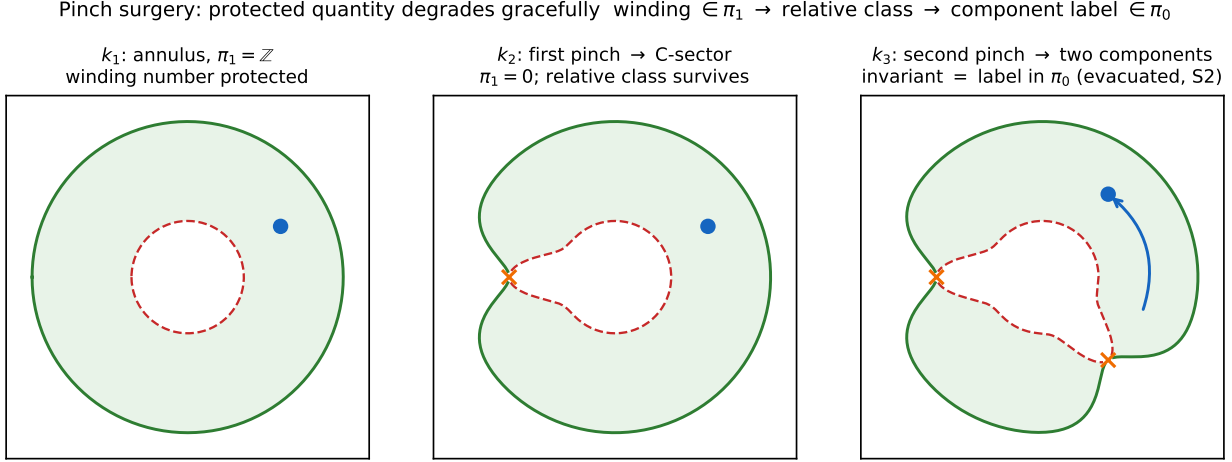


Figure 2: Pinch surgery in three acts. k_1 : intact annulus, $\pi_1 = \mathbb{Z}$, winding number protected. k_2 : a single pinch (orange $\times$) already trivializes π_1 — the safe set becomes a contractible C-sector; only relative endpoint classes survive. k_3 : a second pinch disconnects the safe set; the protected invariant is now the component label in π_0 , monotone under S1, and the trajectory (blue) has been evacuated to the surviving component within the S2 lead time (arrow).

Assumption 5 (S1: monotone excision). $P(k) \subseteq P(k+1)$ for all k (pinches never heal). This is the geometric image of monotone cumulative irreversibility: it holds whenever $r_{\min}$ is nondecreasing in L_{cum} and L_{cum} is nondecreasing in k . Under S1, the component label $j(x_k) \in J(k)$ is a *monotone invariant*: components may split but never re-merge.

Assumption 6 (S2: lead time / evacuation). There is a prediction horizon $\tau \in \mathbb{N}$ and an announced set $\hat{P}(k) \supseteq P(k+\tau)$ available at step k , such that from any $x \in \mathcal{S}(k)$ lying in a *doomed* region (one that $\hat{P}(k)$ will excise or isolate), a surviving component is reachable within τ steps using controls in A^{safe} . A sufficient condition in the toy model: guaranteed tangential authority $\underline{v}_\theta > 0$ throughout the band and $\tau \underline{v}_\theta \Delta t \geq \pi r_{\max}^{\text{sup}}$ (worst-case half-circumference evacuation).

Theorem 2 (Safety across surgeries). *Suppose A1, A3, A4 hold globally, A2 holds on each component $\mathcal{S}_j(k)$ away from an ε_p -collar of its pinch endpoints, and S1–S2 hold. Then for all $k \geq 0$: (i) $x_k \in \mathcal{S}^{\text{surg}}(k)$; (ii) at every surgery time the state lies in a surviving component, and its component label is monotone (irreversibility is realized as topological, not merely metric, exclusion: once a component is severed, no admissible control sequence re-enters it); (iii) within each component, the continuity and relative-homotopy statements of Theorem 1 hold verbatim.*

Proof sketch. Between surgeries, apply Theorem 1 componentwise (A2 holds there). At an announced surgery, S2 provides an admissible evacuation plan whose existence is certified τ steps ahead; executing any element of A^{safe} consistent with the announced excision (i.e., treating $\hat{P}$ -collars

as already forbidden, which only shrinks the fiber and preserves feasibility by the S2 reachability clause) places x in a surviving component at surgery time. Monotonicity of the label follows from S1: excised collars never reopen, and A^{safe} never contains a control whose nominal successor lies in an excised collar. $\square$

Remark 2 (Morse-theoretic reading). View $W(\cdot, k)$ as a time-indexed function on S^1 . A pinch is the event $\min_{\theta} W \downarrow \varepsilon_p$: generically a nondegenerate minimum crossing the threshold, i.e., a saddle-node of the sublevel-set structure of the barrier h . Each surgery is a handle detachment; the sequence of (π_0, π_1) of $\mathcal{S}(k)$ is a barcode-like signature of the system’s irreversibility history, and can be monitored online by persistent homology of $h(\cdot, k)$.

The announcement required by S2 need not be an oracle; A1 makes it *computable from the current geometry*:

Lemma 2 (Constructive lead-time certificate). *Under A1, define the announced set $\hat{P}(k) := \{\theta : W(\theta, k) \leq \varepsilon_p + 2\nu\tau\Delta t\}$. Then $\hat{P}(k) \supseteq P(k+\tau)$.*

Proof. Each boundary drifts at most $\nu\Delta t$ per step, so $W(\theta, k+\tau) \geq W(\theta, k) - 2\nu\tau\Delta t$. If $\theta \in P(k+\tau)$ then $W(\theta, k+\tau) \leq \varepsilon_p$, whence $W(\theta, k) \leq \varepsilon_p + 2\nu\tau\Delta t$. $\square$

Online monitoring via persistent homology. Consider the sublevel filtration $\{\theta : W(\theta, k) \leq s\}$ of the width function on S^1 . Its 0-dimensional persistence diagram is born at the local minima of $W(\cdot, k)$; tracking these minima over k yields pinch *precursors* (low, long-lived minima drifting toward ε_p), and a pinch event is precisely a persistence pair whose birth value crosses ε_p . Lemma 2 converts the monitor into a certificate: any minimum still above $\varepsilon_p + 2\nu\tau\Delta t$ provably cannot pinch within τ steps.

2 When Is the Protection Actually Topological? Codimension-2 Structure

The slogan “the trajectory cannot jump the hole” is a statement about π_1 of the safe set’s complement, and it is *dimension-sensitive*. This section states precisely when the donut’s exclusion is topological and gives a constructive recipe for retaining it in high dimensions. Readers from condensed matter will recognize the logic: it is the same reason vortices are stable in the XY model in 2D (point defects, codimension 2) and dislocation *lines* are stable in 3D, while a puncture of codimension ≥ 3 pins nothing—defects of codimension c are classified by π_{c-1} of the order-parameter space, and a 1D worldline is obstructed only by π_1 .

2.1 The obstruction lives in π_1 of the complement

Proposition 1. *Let $\Sigma \subset \mathcal{X}$ be a closed embedded submanifold of codimension c in a smooth n -manifold, and let the forbidden core be Σ itself (the donut hole being a tubular collar around it).*

1. *If $c = 2$ and the normal bundle of Σ is orientable, small loops linking Σ generate a $\mathbb{Z}$ factor of $\pi_1(\mathcal{X} \setminus \Sigma)$ (locally, the linking number); paths with fixed endpoints fall into distinct homotopy classes indexed by winding, and no continuous deformation avoiding Σ connects them. This is the only codimension in which path-level exclusion is topological. (If the normal bundle is non-orientable, one obtains a $\mathbb{Z}_2$ linking invariant—the annular analogue of $\mathbb{Z}_2$ vortices in nematics, and the natural home for the Kitaev-chain-flavored $\mathbb{Z}_2$ anomaly discussed in the parent DRG note.)*

2. If $c \geq 3$, the complement of Σ inside its tubular neighborhood has π_1 isomorphic to $\pi_1(\Sigma)$ —the linking circle bounds a disk in the normal sphere S^{c-1} , which is simply connected. In $\mathcal{X} = \mathbb{R}^n$ the complement of a $\text{codim} \geq 3$ spine is simply connected: any two trajectories with common endpoints are homotopic avoiding Σ , and “cannot jump the hole” has no topological content; only the metric penalties (G -warping, fiber shrinkage) remain.
3. If the forbidden region is full-dimensional (a thick blob B , boundary of codimension 1), then for $n \geq 3$ the complement $\mathbb{R}^n \setminus B$ is again simply connected. The surviving invariant is the degree of enclosing $(n-1)$ -spheres in $\pi_{n-1}(\mathbb{R}^n \setminus B) \cong \mathbb{Z}$ —it constrains membranes, not paths, and is therefore useless for trajectory-level safety.

The 2D toy model is thus not an innocent illustration: it is the *generic* case only in the plane. In a d -dimensional latent or state space with a low-dimensional spine, claims 8.2 of the parent note must be weakened: the invariance and continuity mechanisms lift, the *topological exclusion does not*, unless one engineers codimension 2.

2.2 A constructive recipe: order-parameter reduction

Topological protection can be *manufactured* in any dimension by choosing the hole as the zero set of a two-component order parameter.

Definition 2 (Safety order parameter). Choose two independent scalar risk functionals and assemble $\psi = (\psi_1, \psi_2) : \mathcal{X} \rightarrow \mathbb{R}^2 \cong \mathbb{C}$, transverse to 0. Examples: ψ_1 = signed margin to the irreversibility threshold ($L_{\max} - \hat{L}$), ψ_2 = signed alignment of exposure (E -projection minus a tolerance). Define the forbidden core $\Sigma := \psi^{-1}(0)$ — automatically a codimension-2 submanifold — and the donut as $\{\rho_{\min} \leq |\psi(x)| \leq \rho_{\max}\}$ in the ψ -plane, pulled back to $\mathcal{X}$.

Proposition 2. *The winding number of $\psi \circ \gamma$ around $0 \in \mathbb{C}$ is a homotopy invariant of trajectories γ avoiding Σ , in any ambient dimension. The generalized filter—radial clip in the ψ -plane, $F_{\text{iso}}(x)$ moving x along a chosen lift of the radial direction $\nabla|\psi|$ —preserves it, by the argument of Theorem 1(4) applied in the plane.*

In physics language: safety carries a $U(1)$ order parameter; danger is a vortex core; trajectories carry a conserved winding charge; pinching a sector is locally gapping out the phase. This gives the “donut” construction a precise meaning in arbitrary dimension and makes the design question concrete: *which two risk signals deserve to be the components of ψ ?* The answer determines what the topology actually protects.

2.3 Reach and well-posedness of the tube

Independently of topology, the tubular construction is well-posed only within the *reach* of the spine (Federer): $\text{dist}(\cdot, \Sigma(t))$ is smooth with unique metric projection only for $r < \text{reach}(\Sigma(t))$, i.e., before hitting the medial axis. We therefore add to A1 the standing requirement

$$\sup_{x, \theta} r_{\max}(\theta, t) < \text{reach}(\Sigma(t)) \quad \text{for all } t,$$

and note that for a *learned* spine this is a nontrivial regularity constraint (high-curvature or self-approaching spines shrink the admissible wall thickness), which should enter the training objective of the learnable- Σ agenda of Sec. 7.

3 Relation to Prior Work

The donut framework intersects several mature literatures; its contribution is best stated as a specific *combination* rather than novelty of any single ingredient. (Citations below are indicative and should be finalized against the standard references.)

Hamilton–Jacobi reachability. HJ methods (Mitchell–Bayen–Tomlin; survey: Bansal et al.) compute maximal safe sets and optimal safe controllers for general nonlinear dynamics with adversarial disturbance, at exponential cost in dimension. The donut fixes an *ansatz* (annular tube around a spine) and buys tractability plus interpretability at the price of conservatism; (2) plays the role that the HJ value function’s temporal derivative bound plays there.

Control barrier functions. CBF-QP filters (Ames et al.; robust ISSf-CBF variants for bounded disturbance) enforce $\dot{h} + \alpha(h) \geq 0$ pointwise. Our A4 is a discrete robust CBF condition; the additions are (i) two coupled barriers with risk-driven motion, (ii) the feasibility condition (2), which standard CBF treatments assume implicitly via a valid-CBF hypothesis, and (iii) the homotopy statement, which pointwise QP filters do not provide—in nonconvex safe sets a QP filter can discontinuously switch which side of an obstacle it projects to, precisely the “jump over the hole” the isotopy-style retraction forbids.

Tube MPC and funnels. Robust tube MPC (Mayne, Langson et al.) and funnel/LQR-tree libraries (Tedrake; Majumdar) also confine trajectories to tubes around nominal plans. The donut differs in that the tube geometry itself is the *risk interface*—radii, metric, and fibers are functions of $\Xi(t)$ —and in the annular (rather than contractible) cross-section, which is what makes topological exclusion expressible at all.

Viability theory. The invariance content of Theorem 1 is a time-varying viability kernel statement (Aubin); S1–S2 add a controlled-surgery layer that, to our knowledge, viability theory does not treat as topology change.

Predictive safety filters and shielding. Learning-compatible safety filters (Wabersich–Zeilinger; Hsu–Fisac et al. survey) and RL shielding (Alshiekh et al.) also modify proposed actions minimally. The donut’s deltas: continuity of the filter as a *trajectory-level* map (Theorem 1(3)), and irreversibility as monotone *topological excision* (Theorem 2(ii)) rather than a penalty or an action mask without geometric semantics.

Topology-aware planning and navigation functions. Homotopy-class constrained planning (h -signature methods, Bhattacharya et al.) and Koditschek–Rimon navigation functions on sphere-worlds already exploit π_1 obstructions; Sec. 2 imports their central lesson (codimension 2 or a quotient construction) into the safety-filter setting and adds the time-varying, risk-driven, surgery-capable annulus.

Honest positioning. Relative to HJ the guarantees here are conditional on the *ansatz* and on (2); relative to CBFs the novelty is topological rather than analytic; the empirical evidence is a 2D toy model. The defensible claim: *a time-varying annular viability tube equipped with (a) a homotopy-class-preserving trajectory filter, (b) monotone topological excision for irreversibility,*

and (c) *risk-warped metric and fibers, with recursive feasibility made explicit*—and, via the order-parameter reduction, a principled route to high dimensions.

Method	Safe-set form	Time-varying	Filter continuity	Topology-aware	Irreversibility
HJ reachability	value sublevel	yes	—	no	via target sets
CBF / ISSf-CBF	$h \geq 0$ blob	rarely	pointwise only	no	no
Tube MPC / funnels	contractible tube	plan-indexed	yes	no	no
Safety filter / shield	set or automaton	sometimes	not guaranteed	no	action mask
h -signature planning	static free space	no	(planning)	yes	no
Dynamic donut	annular tube	risk-driven	trajectory-level	π_1 , surgery	monotone excision

Table 1: Positioning of the dynamic donut among safety frameworks.

4 Stochastic Wind: Probabilistic Invariance

When the exposure wind is a stochastic process rather than a bounded adversary, hard invariance is the wrong target; the natural statements are finite-horizon high-probability invariance and infinite-horizon supermartingale bounds.

Assumption 7 (W1). $\{w_k\}$ are independent, zero-mean, and componentwise sub-Gaussian with proxy σ (bounded noise with $\bar{w} = \sigma$ is a special case).

Proposition 3 (Finite-horizon chance-constrained invariance). *Run the controller of Theorem 1 with the robust margin replaced by the confidence margin*

$$\eta(\delta) = \sigma \Delta t \sqrt{2 \log(4K/\delta)},$$

and with (2) holding with $2\bar{w}$ replaced by $2\eta(\delta)/\Delta t$. If $x_0 \in \mathcal{S}_{\eta(\delta)}(0)$, then

$$\mathbb{P}(x_k \in \mathcal{S}(k) \quad \forall k \leq K) \geq 1 - \delta.$$

Proof sketch. Per step, the deviation of the true successor from the nominal one is sub-Gaussian with proxy $\sigma\Delta t$ in each of the two coordinates; the Chernoff bound gives $\mathbb{P}(\text{deviation} > \eta(\delta)) \leq \delta/(2K)$ per coordinate per step. On the complement event the induction of Theorem 1 goes through verbatim; a union bound over $2K$ coordinate–step pairs yields the claim. $\square$

Proposition 4 (Infinite-horizon supermartingale bound). *Let $d(x, k) = \min\{r - r_{\min}(\theta, k), r_{\max}(\theta, k) - r\}$ be the margin and $B(x, k) = e^{-\lambda d(x, k)}$. Suppose the executed policy guarantees the one-step drift condition $\mathbb{E}[B(x_{k+1}, k+1) \mid x_k] \leq B(x_k, k)$ whenever $d(x_k, k) > 0$ (achievable under W1 when the nominal margin gain of the boundary-repelling control exceeds $\lambda\sigma^2\Delta t^2/2$ plus the boundary drift — a Chernoff computation). Then B is a nonnegative supermartingale and, by Ville’s inequality,*

$$\mathbb{P}(\exists k : x_k \notin \mathcal{S}(k)) \leq B(x_0, 0) = e^{-\lambda d(x_0, 0)}.$$

This is the donut instance of stochastic barrier certificates (Kushner; Prajna et al.): the exit probability decays *exponentially* in the initial margin, which quantifies the value of starting near the spine. Pinch surgery composes with both statements componentwise, since S1–S2 are geometric conditions independent of the noise model (the evacuation clause in S2 should then be read in the same high-probability sense).

5 The Slow Loop $M(t)$: Two-Timescale Stability, Goodhart Robustness, and an Adiabatic Design Rule

Sec. 3.5 closes a feedback loop — effort $\rightarrow$ memory $M \rightarrow$ wall thickness $\rightarrow$ effort — without analyzing it. We model it explicitly as a slow-fast system and give (i) a stability condition via averaging, (ii) a Goodhart-robustness design, and (iii) an adiabatic invariant that answers “how slow is slow enough?” quantitatively.

5.1 Two-timescale model and averaged stability

Let the slow update be

$$M_{k+1} = M_k + \epsilon(\varphi(e_k) - M_k), \quad 0 < \epsilon \ll 1, \quad (3)$$

where e_k is the fast-timescale effort signal (deformation magnitude, near-boundary frequency) and φ is monotone increasing. Thickness responds as $W = W_0 + \kappa_M M$ (high effort $\Rightarrow$ thicker wall, per Sec. 3.5).

Assumption 8 (F1: fast mixing). At frozen M , the closed-loop fast dynamics inside the annulus are geometrically ergodic with mixing time $T_{\text{mix}}(M)$ and stationary mean effort $\bar{e}(M)$.

Proposition 5 (Averaging and stability of the thickness loop). *Suppose F1 and the timescale separation $\epsilon T_{\text{mix}} \ll 1$. Then M_k tracks the averaged dynamics $\dot{M} = g(M) := \varphi(\bar{e}(M)) - M$ with error $O(\epsilon T_{\text{mix}})$ over times $O(1/\epsilon)$, and a fixed point M^* is exponentially stable iff*

$$\left. \frac{d}{dM} \varphi(\bar{e}(M)) \right|_{M^*} < 1.$$

In particular, if the natural monotonicity holds — more room $\Rightarrow$ less deformation effort, i.e. $\bar{e}'(M) \leq 0$ — then $(\varphi \circ \bar{e})' \leq 0$, $g'(M^) \leq -1$, and the averaged loop is unconditionally stable with no oscillation.*

Remark 3 (Failure modes). The two ways this breaks are instructive. (i) *Delay-induced oscillation*: when $\epsilon T_{\text{mix}} = O(1)$ the fast subsystem has not equilibrated between slow updates; the effective loop acquires a delay and can enter relaxation oscillations (breathing donut). The fix is structural: enforce $\epsilon \leq c/T_{\text{mix}}$, or add a hysteresis deadband to (3). (ii) *Non-monotone effort response*: if $\bar{e}(M)$ is non-monotone (e.g., a thicker wall invites riskier exploration that *raises* effort), $(\varphi \circ \bar{e})'$ can exceed 1 locally and the loop self-excites. This is detectable offline by estimating $\bar{e}(\cdot)$ over the operating range of M .

Crucially, the slow loop must also respect the viability rate condition (2): since M moves the boundaries at rate $\leq \epsilon \kappa_M / \Delta t$, we simply *include this in the budget*,

$$\nu \geq \nu_{\text{risk}} + \epsilon \kappa_M / \Delta t,$$

so that by construction no excursion of the slow loop — stable, oscillating, or adversarially driven — can ever outrun the control authority.

5.2 Goodhart robustness of the effort signal

The effort e_k is agent-influenced: an agent that wants more room can manufacture near-boundary events to inflate e and fatten the wall; one that wants a lax future audit can feign calm to thin it and exploit later. Three design rules make (3) robust to such strategies:

1. **Asymmetric rates and evidence.** Thinning (the conservative direction, a *smaller* safe set) may be fast and unaudited; thickening (granting freedom) is rate-limited and requires *verifier-side* evidence — e.g., boundary-distance quantiles computed from a trusted state estimate, never from agent-reported statistics.
2. **Robust statistics.** Use a trimmed quantile of the margin $d(x_k, k)$ over a window rather than event counts; single manufactured grazings then move e by $O(1/\text{window})$.
3. **Bounded influence.** Cap $M \in [0, M_{\max}]$ with $\epsilon \kappa_M$ inside the ν -budget as above; then even a fully adversarial effort signal degrades performance (over/under-constraint) but can never violate safety — Goodharting the slow loop is a *liveness* problem, not a safety problem.

5.3 An adiabatic invariant: how slow is slow?

There is a sharper, physics-native answer to the modulation-rate question. Inside the band, the tangentially-preferred dynamics make the radial coordinate an effective oscillator confined between the walls; let T_{fast} be its period (of order the circulation time $2\pi r/\bar{v}_\theta$ or the radial bounce time $2W/\bar{v}_r$, whichever is shorter), and define the radial action

$$J = \frac{1}{2\pi} \oint p_r dr$$

over one fast cycle at frozen geometry.

Proposition 6 (Adiabatic invariance of the band action). *Define the adiabaticity parameter*

$$\varepsilon_{\text{ad}} := \sup_{\theta, t} \left| \partial_t \log W(\theta, t) \right| \cdot T_{\text{fast}}.$$

If $\varepsilon_{\text{ad}} \ll 1$, then J is conserved to $O(\varepsilon_{\text{ad}})$ over times $O(1/\varepsilon_{\text{ad}})$ (and to exponential accuracy for analytic modulation, Nekhoroshev-type). Consequently the trajectory’s relative depth in the band, $(r - r_{\text{mid}})/W$, is preserved as the donut breathes: the state rides the modulation without being pumped toward either wall.

Conversely, when $\varepsilon_{\text{ad}} = O(1)$ — a sudden squeeze — radial action is parametrically pumped and the trajectory is driven to graze the boundaries *even though every step is individually feasible*; this is precisely the jitter phenomenology of barrier-only control observed in Fig. 5, now with a mechanism (see Fig. 3).

The design rule is then quantitative:

Risk- and M -driven radius changes must satisfy $\varepsilon_{\text{ad}} \lesssim 0.1$; anything faster is not a modulation but a *surgery*, and must be handled by Sec. 1.2.

This cleanly separates the framework’s two regimes — graceful geometric deformation (adiabatic) versus topological events (non-adiabatic) — and gives the slow loop ϵ an operational upper bound: $\epsilon \kappa_M / W \leq 0.1 / T_{\text{fast}}$.

Errata to the Parent Note (apply directly)

1. **Sec. 1, last line:** “It is ‘ ’ (donut) safety” has a dropped glyph. Replace with: “It is *donut* safety: stay in the tasty annular region; the hole is poison, the outside is chaos.”

Adiabatic invariance of the band action: slow modulation preserves J and relative depth;
fast modulation pumps radial action and drives boundary grazing (mechanism of barrier-only jitter)

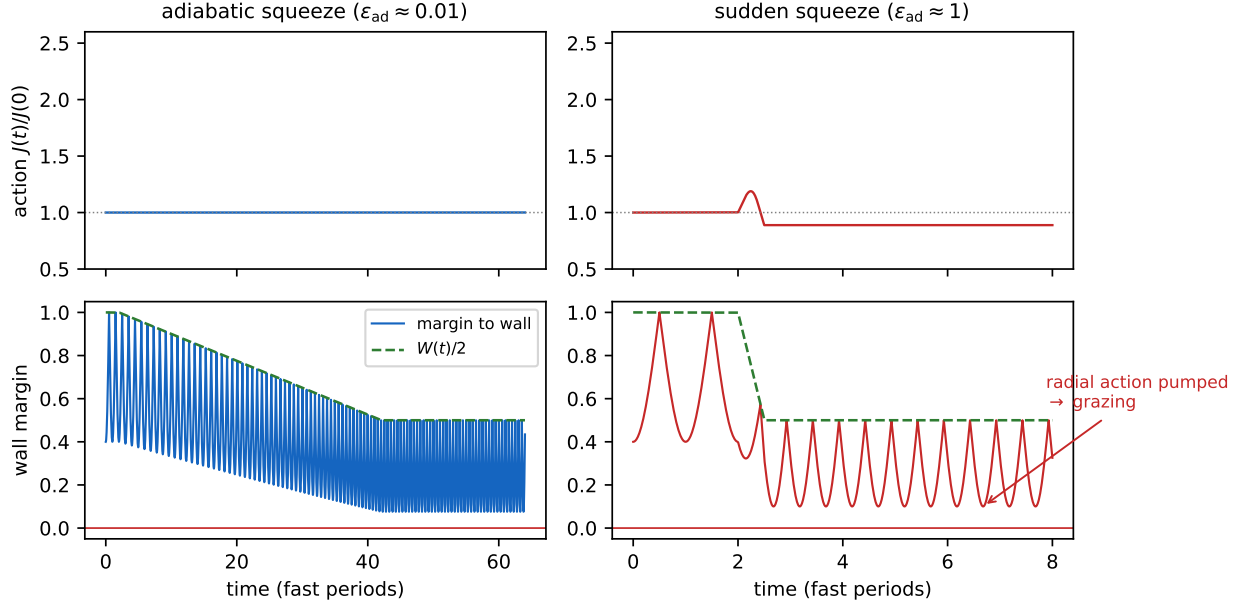


Figure 3: Numerical check of Proposition 6 on the radial oscillator $\ddot{x} = -\omega(t)^2 x$, $\omega \propto 1/W$, with the wall width $W(t)$ ramped down by 50%. Left column (slow ramp over ~ 40 fast periods): the action $J = E/\omega$ is conserved to $\sim 1\%$ and the wall margin shrinks smoothly with $W/2$. Right column (same ramp over half a fast period): J is pumped by a factor ~ 2 and the margin collapses toward grazing — the geometric mechanism behind barrier-only jitter.

2. **Ξ vs. Φ :** $\Xi(t)$ is a *tuple* of risk channels while $\Phi(t)$ is a *scalar aggregate*; the relation is never stated. Insert after the definition of Φ : “ $\Phi = \Lambda(\Xi)$ is a fixed monotone aggregation (here: the weighted sum above) of the risk tuple Ξ ; all geometric moduli depend on Ξ only through Φ and L_{cum} unless stated otherwise.”
3. **Directional inconsistency (O/U):** Sec. 3.3 says the fiber shrinks with “falling O/U” while Sec. 5 uses $R_{OU} = U/(O+\epsilon)$ (rising). Fix by defining O (observability margin) and U (uncertainty / uncontrollability) explicitly once, using the single ratio $R_{OU} = U/(O+\epsilon)$ throughout, and rewording Sec. 3.3 to “with rising R_{OU} ”. (Also reconcile the label “controllability-to-observability ratio” in Sec. 5 with whichever semantics is chosen.)
4. **Sec. 8.1:** “Figure 6” should be “Figure 10”.
5. **Fig. 8 caption:** “Final L (full): 0.00” contradicts the claim that the inner hole enlarged due to accumulated L . Either report the F_{iso} -only run’s final L (the run that actually accumulates irreversibility) alongside the full controller’s, or rerun with parameters producing nonzero final L for the full controller; amend the caption to attribute the hole expansion to the correct trajectory.
6. **Undefined symbol:** $\mathcal{I}_{\text{constraint}}$ in “ $\mathcal{R}(X_0; \mathcal{I}_{\text{constraint}}) \cap X_{\text{forbid}} = \emptyset$ ” is never defined. Either define it (“the constraint interface, i.e., the tuple $(G, A_x, h, F_{\text{iso}}, M)$ acting on admissible

dynamics”) at first use, or replace the clause with “the forward reachable set under the constrained dynamics is disjoint from X_{forbid} ”.